MIL-F-80146A
13 February 1974
SUPERSEDING
MIL-F-80146
27 March 1968

MILITARY SPECIFICATION

FURNACES, HEAT TREATING, GAS FIRED, PIT TYPE, FORCED CONVECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vertical, pit type, heat treating furnaces of the gas fired, forced convection design.

1.2 Classification. The furnaces shall be of the following classes as specified (see 6.2.1).

Class 1 - 350 - 1250° F Operating Temperature

Class 2 - 600 - 1750° F Operating Temperature

* 1.2.1 Size. The size of the furnace work chamber shall be as specified (see 6.2.1), and should be selected from the sizes listed in Table I. The dimensions specified shall be the minimum inside effective working dimensions of the heating chamber in which performance requirements and temperature uniformity specified herein shall be maintained.

TABLE I

SIZE, FURNACE WORKING (Dimensions in Inches)					
Diameter	Depth	Diameter	Depth	Diameter	Depth
25	24	33	60	38	72
25	30	35	36	38	84
25	36	35	48	48	48
25	48	35	60	48	60
25	60	38	36	48	72
33	36	38	48	48	84
33	48	38	60	60	60

FSC 3424

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2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-F-3296	Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking), Packaging of.

STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments.	Occupational Safety and Health Administration Standards

NATIONAL BUREAU OF STANDARDS (NBS) HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

* JOINT INDUSTRIAL COUNCIL (JIC)

EGP - 1 Electrical Standards for General Purpose Machine Tools

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101.)

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* FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Handbook of Industrial Loss Prevention, Chapter 37

(Application for copies should be addressed to Factory Mutual Engineering Corporation, 1151 Boston - Providence Turnpike, Norwood, Mass. 02062.)

* AMERICAN NATIONAL STANDARDS

ANS C96.1 Temperature Measurement Thermocouples

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association standards and specifications are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

* 3.1 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract. For single item procurements, the furnace submitted to fulfill the terms of the contract or purchase order shall be considered the First Article.

3.2 Design. Furnaces and equipment provided in accordance with this specification shall be new and of the manufacturer's current design. The furnace shall be of the vertical pit, gas fired, forced convection type with a cylindrical work chamber. The opening shall be located in the top of the furnace for inserting and removing the charge in a vertical plane. The furnace shall consist of two chambers: the heating chamber and the work chamber. The work chamber shall be insulated from the heating chamber to prevent direct radiation on the work. The work chamber shall be heated by 100 percent forced convection. The heated gasses shall be transferred from the heating chamber to the work by means of a circulating fan. The fan shall be located outside the work chamber. Each furnace shall be furnished complete with the necessary burners, blowers, gas-air mixers, pressure regulators, valves, pilots, manifolds, and piping. The combustion system shall be of the automatically operating proportioning design, capable of being adjusted to produce a furnace atmosphere from rich to neutral to lean. The system shall maintain the preset ratio over the proportioning

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range. Unless otherwise specified, the furnace shall be designed to operate on a fuel supply of natural gas with a nominal heat content of 1,000 BTU per cu ft at a supply pressure from 6 to 8 inches water column. When required, the furnace shall be designed and equipped to operate on either a butane or a propane fuel supply as specified (see 6.2.1).

* 3.2.1 Controlled Atmosphere. When required, the furnace shall be designed for, and provided with, a retort for operation with a controlled atmosphere. Retort requirements, retort fan if required, location of fan, either in top or bottom of retort, and the type of atmosphere that will be used in the retort, shall be as specified (see 6.2.1). The dimensions specified for the size furnace shall be the minimum inside useful working dimensions of the retort. The controlled atmosphere furnace may be of the single chamber design, with burners firing directly into the work chamber. The flame of the burners shall not impinge directly on the retort, and the temperature uniformity specified herein shall be maintained in the retort.

* 3.2.2 Safety requirements. Each furnace shall be provided with safety features and devices for the protection of personnel and equipment. In addition to other safety requirements specified herein, the safety system of the furnace shall conform to the requirements of FM "Handbook of Industrial Loss Prevention", Chapter 37. Except as otherwise specified by the procuring activity, the furnace shall comply with the standards promulgated under OSHA applicable to the furnace manufacturer. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

* 3.2.3 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating lubrication system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.4 Material. All material used in the fabrication of the furnace shall be new and of the quality necessary to produce a furnace to meet the requirements described herein. Materials that are exposed to the atmosphere of the heating chamber shall be compatible with the type of atmosphere to be used in the furnace.

3.2.5 Accessibility. All parts subject to wear, breakage, or distortion, and all parts which require periodic maintenance, shall be accessible for adjustment and replacement, as applicable.

* 3.2.6 Maintainability. Item configuration and adjustment features will be designed to reduce maintenance operations to the simplest procedures compatible with a minimum of maintenance skills and common maintenance equipment. Instructions for disassembly, repair, adjustment, and reassembly will be concise and clearly defined, in operation and maintenace manuals.

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3.2.7 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or furnace.

3.3 Construction. The furnaces shall be constructed of parts which are new and without defects. Repaired parts shall not be used. The structures shall be capable of withstanding all forces encountered during operation of the furnace to their maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, and soldering shall be employed only where required in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

* 3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.5 Painting. The furnace shall be primed and painted with heat resisting paint in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to NBS Handbook H28.

* 3.3.7 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

* 3.4 Electrical equipment. The furnace shall be furnished complete with all accessory electrical motors, controls, relays, signal lights, and switches pre-wired where possible for operation upon connection to source of power. All power actuated, manual adjusted, and automatic controls shall be designed for operation on nominal 115 volt, 60 cycle current. A fusible disconnect switch shall be provided to interrupt all electrical power to the furnace. All electrical equipment shall conform to JIC Electrical Standard for General Purpose Machine Tools.

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* 3.4.1 Motors. Unless otherwise specified (see 6.2.1), motors rated in excess of 1/4 horsepower shall be of the dual voltage 230/460 volt, 60 HZ, three phase type, wired to operate on 440 volt power supply.

* 3.4.2 Electromagnetic interference control. When specified (see 6.2.1), the furnace electrical system shall comply with the requirements of MIL-STD-461 for Class IIB equipment.

3.5 Performance. Each furnace shall be capable of continuous operation at any temperature within its operating range with maximum loads as specified in Table II.

* 3.5.1 Recovery Rate. The recovery rate shall be not less than that specified in Table II. The recovery rate is based on gross load, and includes the weight of baskets, trays or fixtures required to retain the workload. The recovery rate is determined by loading the specified load in the furnace which has been stabilized at the temperature shown in Table II. If the furnace is required to operate with a controlled atmosphere using a retort, or if a size other than the sizes listed in Table II is required, the recovery rate and maximum load shall be as specified (see 6.2.1).

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TABLE II

RECOVERY RATE AND MAXIMUM LOAD
(not less than)

WORK CHAMBER DIMENSIONS (in inches)		CLASS 1 Recovery Rate, Lbs steel/hr to 1200° F	CLASS 2 Recovery Rate, Lbs steel/hr to 1700° F	MAXIMUM LOAD, POUNDS
Diameter	Depth			
25	24	700	600	2000
25	30	750	650	2500
25	36	800	700	3000
25	48	850	750	3000
25	60	900	800	3000
33	36	850	750	4000
33	48	900	800	4500
33	60	950	900	4500
35	36	900	750	4500
35	48	950	850	4500
35	60	1000	900	4500
38	36	900	800	4500
38	48	950	900	4500
38	60	1050	950	4500
38	72	1100	1000	5000
38	84	1150	1050	5000
48	48	1000	950	6000
48	60	1100	1000	6500
48	72	1200	1100	7000
48	84	1300	1200	8000
60	60	1200	1100	8000

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* 3.5.2 Temperature uniformity. With the furnace charged and the temperature stabilized, the temperature variation from chamber control point to any other point within the chamber working area shall not vary more than ± 10° F for Class 1, or ± 15° F for Class 2 furnace. Each furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and at any desired temperature control point within the operating temperature range specified for the class furnace.

3.6 Components

* 3.6.1 Shell. The outer shell of the furnace shall be fabricated from sheet steel of at least 3/16 inch in thickness, and properly reinforced with steel members and shapes. Where joining by welding is required, only electric welding with continuous welds shall be used. The top and bottom plates shall be fabricated from steel plate at least 3/8 inch thick.

3.6.2 Lining. The lining of the heating chamber and work chamber shall consist of at least 2000° F insulating fire brick, backed up with insulating slab or preformed insulation. Overall thickness and quality of the lining shall be such that the outside surface temperature of the Class 1 furnace will not exceed 100° F above ambient, or the Class 2 furnace 150° F above ambient, after the furnace temperature has stabilized at maximum operating temperature. Powdered, shredded, or granular material shall not be used for insulating purposes. Means shall be provided for protection of the lining wall from damage during insertion and removal of the work container or charge.

* 3.6.3 Cover. The furnace shall be provided with an insulated cover. The insulation may be refractory, block or bulk type insulation. The quality and thickness of the insulation shall be such that the surface temperature of the cover will be compatible with the outside surface temperature of the shell. The cover shall rest on an asbestos composition gasket. The gasket shall be retained in a channel in the shell or insulation. Furnaces designed to operate with a carburizing atmosphere shall be provided with an ore seal trough. Unless otherwise specified, the cover shall be provided with a mechanical mechanism for easy lifting, and the cover shall swing horizontally, from left to right, from the closed position when facing the furnace. The maximum effort required to lift and swing or close the cover shall not exceed a force of 40 pounds. When required, the cover shall be power operated or designed for hoist removal as specified (see 6.2.1).

3.6.4 Work container. The furnace shall be provided with a solid sidewall work container with inside dimensions of not less than the dimensions specified for the size furnace. The container shall be cast or fabricated from nickel chromium alloy. The bottom shall be either cast or fabricated grid, and shall include a wire mesh screen or perforated metal plate on top of

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the bottom grid. The screen shall have sufficient open area to allow adequate circulation through the work area. The container shall be designed and constructed so that when it is seated in the chamber, all gases will pass through the work area. The container shall be capable of supporting maximum loads from the bottom grid or from a spider type fixture supported on the top of the container. When specified (see 6.2.1) the container shall consist of individual trays capable of being stacked one on top of the other. The trays shall be provided with removable bottoms and means for locking together for handling as a single container. Tray depths, or the number required to make up a container, shall be as specified. When required, trays or baskets that will fit inside the container shall be provided as specified. All work containers and lifting devices shall be provided with lifting eyes, rings, or trunnions, capable of handling maximum loads.

3.6.5 Circulating system. Circulation of the heated gases within the furnace shall be accomplished by a nickel-chromium alloy circulating fan. The capacity of the circulating system shall be sufficient to produce and maintain the temperature uniformity specified in 3.5 for each class furnace. The direction of the gas flow shall be from the top to bottom of the work chamber. Fan bearings shall be located outside the heated area and protected from the radiant heat of the shaft. The system shall provide for automatically shutting off the heat system, if the fan stops for any reason.

3.6.6 Thermocouples. Chromel-alumel type thermocouples conforming to ANS C96.1 shall be used as the temperature sensing device for each instrument. The thermocouples shall be protected from the contamination of the furnace gases by protective tubes. Thermocouple lead wires shall be at least 25 feet in length and of the same material as the thermocouples, or of a material having the same thermoelectric characteristics at a thermocouple head temperature between 0 degrees and 250° F.

* 3.6.7 Program control. When required, the furnace shall be provided with a program control system integrated with the instrument control system. The programming requirements, such as heating rates, soak periods and cooling rates, shall be as specified (see 6.2.1).

* 3.6.8 Instruments. Each furnace shall be provided with a temperature indicating-recording-controller type instrument with an accuracy of 0.25 percent of full scale reading. Solid state design shall be employed where possible. The instruments shall be marked to indicate the particular type of thermocouple for which it is calibrated. Automatic reference junction compensation shall be provided in all instruments using thermocouples as the primary temperature sensing element.

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* 3.6.8.1 Temperature recorder. The temperature recorder shall be of the strip chart, self-balancing type. The recorder shall include a safety switch operated from the front of the instrument that may be positioned above set point to shut off heat under excess temperature condition. The instrument shall also be provided with a thermocouple upscale device which will act through the safety system to shut off the heat source in the event of thermocouple failure. Unless otherwise specified, the calibrated chart width shall be not less than 6 inches, and the chart length shall be not less than 80 feet to the roll. The chart speed shall be one inch per hour. The chart mechanism shall be furnished in a dust-proof housing with a glass door that permits full view of chart width and not less than 7 inches of chart length. When required, the chart width shall be at least 9 or 12 inches, as specified (see 6.2.1).

* 3.6.8.2 Temperature controller. The temperature controller shall be of the proportioning, automatic, controlling type. The controller shall be capable of maintaining the set control temperature with a deviation of not more than $\pm$ 2° F.

* 3.6.8.3 Temperature indicator. The temperature indicator scale shall be approximately the same length as the temperature recorder chart width. The calibrated scale range shall be at least 100° F above the maximum temperature of the furnace. The increments between successive temperature marks shall not be greater than 20° F, with the spacing of numbered marks not greater than 400° F. The control indicator shall be provided with two pointers. One pointer shall be adjustable to the desired control temperature, and one pointer shall indicate the actual furnace temperature.

3.6.8.4 Excess temperature control. The furnace shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the operating control circuitry. The controller shall be of the indicating type with a pointer which may be set at the over-temperature control point, and shall function to shut off the heat source to the furnace when the over-temperature control point is reached. In case the furnace is shut down by the excess temperature control system, the system shall sound an audible alarm, and the furnace shall remain off until started manually. Calibrated accuracy shall be within one percent of full scale reading.

* 3.6.8.5 Instrument panel. Floor mounted instrument panels shall be provided for mounting all instruments and accessory controls. The panels shall conform to JIC Standard EGP-1. All ventilated openings shall be designed to prevent entrance of any deleterious substance. Signal panel lights shall be provided to indicate the furnace cycle performance. The panels shall provide flush or semi-flush mounting of all instruments.

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3.7 Work containers and fixtures. When specified (see 6.2.1) fixtures and extra work containers shall be furnished. Number required, design, and dimensions shall be as specified.

3.8 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

3.9 Nameplate. Unless otherwise specified (see 6.2.1) a corrosion resistant metal nameplate, marked in accordance with MIL-STD-130 shall be securely attached to each furnace. The nameplate shall contain the information listed below. If the furnace is a special model, the model designation shall include the model of the basic standard furnace and a suffix keyed to the manufacturer's permanent records.

Nomenclature

Manufacturer's name

Manufacturer's model number

Manufacturer's serial number

Type of gas

Gas input requirements (CFM)

Electrical input characteristics and rating

Inside work chamber dimensions (diameter and depth)

Operating temperature range (degrees F)

Date of manufacture

Contract or order number

Federal Stock Number or Plant Equipment Code

U S

3.10 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.11 Workmanship. Workmanship of the furnace and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1), First Article inspection shall consist of the examination in 4.5 and all tests in 4.6. Any examination, inspection, or test performed under 4.4 need not be repeated under 4.3. Failure of the item to pass any examination or test shall be cause for disapproval of the First Article.

4.4 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6.1 and the inspection in 4.7. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.5 Examination. The furnace and equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment, safety conformance, and workmanship to determine compliance with the requirements of this specification.

* 4.6 Tests. All tests shall be conducted at an ambient temperature of 77° F ± 18° F. Test instruments and sensing elements, used to perform the test specified for the furnace and the instruments should have been checked against a standard potentiometer type instrument and calibrated thermocouple of known accuracy, within the previous three months to read accurately within ± 2° F.

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* 4.6.1 Instrument test. Indicating and recording instruments shall be tested by impressing various predetermined voltages, simulating thermocouple outputs for different furnace temperatures to terminals where normally the thermocouples would be connected. These tests shall be made in accordance with, and results checked against, standards established in ANS C96.1.

4.6.2 Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance, and faulty grounds, in accordance with the requirements of JIC Standard EGP-1.

* 4.6.3 Electromagnetic interference control test. Equipment having electromagnetic interference control shall be tested for compliance with 3.4.2 in accordance with Table II of MIL-STD-461 for Class IIB equipment, test methods CE03 and RE02.

* 4.6.4 Operational test. The furnace shall be prepared for operation, and operated in accordance with manufacturer's handbook of instructions. The furnace shall be operated empty at 600° F (Class 1) and at 1200° F (Class 2) to check operation of control system and fan rotation direction. The fan interlock and all safety devices shall be tested by simulating failures that normally would cause the device or system to operate. Each system shall perform its intended function. The cover lift mechanism shall be checked, and the effort required to lift, swing, or close the cover shall not exceed a force of 40 pounds. Upon completion of the above test, the empty furnace shall be operated until the refractory and insulation has cured and the temperature stabilized at the recovery temperatures listed for the class furnace in Table II. After equilibrium has been established, the furnace shall be loaded with a charge of iron or steel equivalent to at least the recovery rate specified in Table II. At the end of one hour, the chamber temperature shall have recovered and stabilized at the initial temperature within $\pm$ 2° F. Furnaces equipped for programmed cycles shall be operated through a complete cycle. The presence of leaks, or any damage to the furnace or control equipment, shall be cause for rejection.

* 4.6.5 Performance test. The furnace shall be loaded with a charge of iron or steel equivalent to at least 50% of maximum load specified for the size furnace, and a chamber temperature survey shall be performed. A minimum of nine test locations shall be used, with eight thermocouples located symmetrically through the load so as to scan the work space. One thermocouple shall be located within 6 inches of the control couple. For furnaces with a chamber volume of less than 10 cubic feet, the test may be performed with a minimum of three thermocouples, with one each located in the top, middle and bottom of the chamber. The survey shall be performed at the minimum and at the maximum operating temperature of the furnace. Readings should be continuously recorded or taken frequently enough to determine when the temperature of the test zone approaches the bottom of the temperature range being

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surveyed. From that time until thermal equilibrium is reached, none of the temperature readings should exceed the maximum of the range being surveyed. After thermal equilibrium is reached, readings shall be taken at not greater than 5 minute intervals for at least 30 minutes, and the maximum temperature variation of all points under test shall not exceed the limits specified in 3.5 for the class furnace under test. With the temperature stabilized at maximum operating temperature, the outside surface shell and cover temperature shall be checked at random locations, and shall not exceed the limits specified for the class furnace.

4.7 Inspection of preparation for delivery. Preservation-packaging, packing, and marking for shipment and storage shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging, packing, and marking. Preservation-packaging, packing and marking shall be in accordance with MIL-F-3296. Level of preservation-packaging, and of packing shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. Furnaces covered by this specification are intended primarily for use in such heat treat operations as tempering, annealing, hardening, and normallizing metal parts and materials.

* 6.2 Ordering data.

* 6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size (diameter and depth) required (see 1.2).

c. First Article approval, if required (see 3.1).

d. Type of fuel, if different (see 3.2).

e. Retort for operation with a controlled atmosphere, type of atmosphere, size, and fan location, when required (see 3.2.1).

f. Specify exceptions and safety and health requirements (see 3.2.2 and 6.5).

g. Motor electrical characteristics, if different (see 3.4.1).

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h. Electromagnetic interference control, if required (see 3.4.2).

i. Recovery rate and maximum load, if the size specified is different from sizes listed in Table II, or if a retort is required (see 3.5.1).

j. Power operated or hoist removal cover, if required (see 3.6.3).

k. Work container trays or baskets if required (see 3.6.4) (Specify depth or number required to make up a set).

l. Program control when required, and program requirements (see 3.6.7).

m. Chart width, if different (see 3.6.8.1).

n. Extra work containers or fixtures, if required (see 3.7).

o. Repair parts, if required (see 3.8).

p. Name plate, if different (see 3.9).

q. Quality conformance inspection, if different (see 4.3).

r. First Article inspection, if different (see 4.4).

s. Level of preservation, packaging, and packing required (see 5.1).

* 6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First Article. When First Article approval is required under 3.1, the solicitation should adhere to the applicable portions of Section 1, Part 19 of the Armed Services Procurement Regulation.

* 6.4 Mercury prohibited. When the use of mercury or its compounds in the furnace and its components and accessories is prohibited, procurement documents should include a statement to that effect.

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* 6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the Standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point of operation, and other safety and health requirements should be as specified.

* 6.6 Material for tests. The quantity of material required for operational and performance tests is generally not available at manufacturer's plants. Also, before testing, the refractory and insulation must be completely cured which increases the possibility of damage in movement and transit of the furnace. For these reasons, consideration should be given to performing First Article testing after the furnace has been installed at the site of operation.

* 6.7 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document, based on the entire content, irrespective of the marginal notations and relation to the last previous issue.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Navy - SH
Air Force - 84
DSA - GS

User Activities:
Army - SM, MU, WC
Navy - MC, OS

Preparing Activity:
DSA-IP

Project Number:
3424-0056

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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